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*
*          STAAD.Pro
*          Version 2007    Build 04
*          Proprietary Program of
*          Research Engineers, Intl.
*          Date=    MAY 21, 2013
*          Time=    17:23:58
*
*          USER ID: Parsons Brinckerhoff
*****

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1. STAAD SPACE

INPUT FILE: vent duct VER3.STD

2. START JOB INFORMATION

3. ENGINEER DATE 27-NOV-12

4. END JOB INFORMATION

5. INPUT WIDTH 72

6. UNIT METER KN

7. JOINT COORDINATES

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8. 1 0 0 0; 2 3.75 0 0; 3 0 5 0; 4 0 5 3.5; 5 3.75 5 0; 6 3.75 5 3.5
9. 7 0 0 3.5; 8 3.75 0 3.5; 9 0 8 0; 10 0 8 3.5; 11 3.75 8 0
10. 12 3.75 8 3.5; 13 0 11 0; 14 0 11 3.5; 15 3.75 11 0; 16 3.75 11 3.5
11. 17 0 14 0; 18 0 14 3.5; 19 3.75 14 0; 20 3.75 14 3.5; 29 0 12.5 0
12. 30 3.75 12.5 0; 31 0 12.5 3.5; 32 3.75 12.5 3.5; 33 3.75 2 0
13. 34 3.75 2 3.5; 35 0 2 0; 36 0 2 3.5; 37 0.5 5 0; 38 0.5 5 3.5
14. 39 0.5 8 0; 40 0.5 8 3.5; 41 0.5 11 0; 42 0.5 11 3.5; 43 0.5 14 0
15. 44 0.5 14 3.5; 45 0.5 2 0; 46 0.5 2 3.5; 47 3.25 5 0; 48 3.25 5 3.5
16. 49 3.25 8 0; 50 3.25 8 3.5; 51 3.25 11 0; 52 3.25 11 3.5; 53 3.25 14 0
17. 54 3.25 14 3.5; 55 3.25 2 0; 56 3.25 2 3.5; 57 0 9 0; 58 3.75 9 0
18. 59 0 9 3.5; 60 3.75 9 3.5

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19. MEMBER INCIDENCES

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20. 1 1 35; 2 2 33; 3 3 37; 4 3 4; 6 7 36; 7 8 34; 8 4 38; 17 3 9; 18 5 11
21. 19 9 39; 20 9 10; 21 11 12; 22 4 10; 23 6 12; 24 10 40; 25 10 3; 26 9 4
22. 29 12 5; 30 11 6; 33 9 57; 34 11 58; 35 13 41; 36 13 14; 37 15 16
23. 38 10 59; 39 12 60; 40 14 42; 41 14 9; 42 13 10; 45 16 11; 46 15 12
24. 49 13 29; 50 15 30; 51 17 43; 52 17 18; 53 19 20; 54 14 31; 55 16 32
25. 56 18 44; 57 18 13; 58 17 14; 61 20 15; 62 19 16; 95 5 6; 104 53 41
26. 105 43 51; 116 29 17; 117 30 19; 118 31 18; 119 32 20; 120 33 5
27. 121 6 33; 122 34 6; 123 5 34; 124 35 3; 126 36 4; 129 35 45; 131 36 46
28. 132 33 34; 133 37 47; 134 38 48; 135 39 49; 136 40 50; 137 41 51
29. 138 42 52; 139 43 53; 140 44 54; 141 45 55; 142 46 56; 143 47 5
30. 144 48 6; 145 49 11; 146 50 12; 147 51 15; 148 52 16; 149 53 19
31. 150 54 20; 151 55 33; 152 56 34; 153 54 42; 154 44 52; 155 51 39
32. 156 41 49; 157 52 40; 158 42 50; 159 49 37; 160 39 47; 161 50 38
33. 162 40 48; 163 47 45; 164 37 55; 165 48 46; 166 38 56; 167 57 13
34. 168 58 15; 169 59 14; 170 60 16

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35. START GROUP DEFINITION

36. JOINT

37. _COLBASE1 1 2 7 8

38. MEMBER

39. _COL1 1 2 6 7 17 18 22 23 33 34 38 39 49 50 54 55 116 TO 120 122 124 -

40. 126 167 TO 170

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41. _LBEAM1 20 21 36 37 52 53
42. _LBEAM1A 4 95 132
43. _LOADBEAM 3 8 19 24 35 40 51 56 129 131 133 TO 152
44. _ANGLE1 104 105 153 TO 166
45. _ANGLE2 25 26 29 30 41 42 45 46 57 58 61 62 121 123
46. _SPLICE 49 50 54 55
47. _LOADBEAM1A 3 8 19 24 35 40 51 56 129 131
48. _LOADBEAM1B 143 TO 152
49. END GROUP DEFINITION
50. MEMBER PROPERTY JAPANESE
51. _COL1 TABLE ST H414X405X18
52. _LBEAM1 TABLE ST H150X150X7
53. _LBEAM1A TABLE ST H150X150X7
54. _LOADBEAM TABLE ST H150X150X7
55. _ANGLE1 TABLE ST C250X90X9
56. _ANGLE2 TABLE ST C250X90X9
57. MEMBER RELEASE
58. _ANGLE1 START MX MY MZ
59. _ANGLE1 END MX MY MZ
60. _ANGLE2 START MX MY MZ
61. _ANGLE2 END MX MY MZ
62. _LBEAM1 START MY MZ
63. _LBEAM1 END MY MZ
64. _LBEAM1A START MY MZ
65. _LBEAM1A END MY MZ
66. _LOADBEAM1A START MZ
67. _LOADBEAM1B END MZ
68. DEFINE MATERIAL START
69. ISOTROPIC MATERIAL1
70. E 2.05E+008
71. POISSON 0.3
72. DENSITY 77
73. ISOTROPIC STEEL
74. E 2.05E+008
75. POISSON 0.3
76. DENSITY 76.8195
77. ALPHA 1.2E-005
78. DAMP 0.03
79. END DEFINE MATERIAL
80. CONSTANTS
81. *BETA 45 MEMB _ANGLE1
82. MATERIAL STEEL ALL
83. SUPPORTS
84. 1 2 7 8 PINNED
85. LOAD 1 DL
86. SELFWEIGHT Y -1
87. LOAD 2 LL
88. MEMBER LOAD
89. _LOADBEAM1A CON GY -2.508 0.35
90. _LOADBEAM1B CON GY -2.508 0.15
91. LOAD 3 WX
92. MEMBER LOAD
93. _COL1 UNI GX 0.283
94. _LBEAM1 UNI GX 0.141
95. _LBEAM1A UNI GX 0.141
96. * WIND LOAD = 0.613 X 34 X 34 = 708.628 N/M2
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STAAD SPACE

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97. * ACTING ON ONE JOINT OF THE VENTILATION DUCT = 3M X 1.5M X 708.628 /1000 = 3.1
98. _LOADBEAM1A CON GX 3.189 0.35
99. _LOADBEAM1B CON GX 3.189 0.15
100. LOAD 4 WZ
101. MEMBER LOAD
102. _COL1 UNI GZ 0.283
103. _LOADBEAM UNI GZ 0.141
104. _LOADBEAM1A CON GZ 3.189 0.35
105. _LOADBEAM1B CON GZ 3.189 0.15
106. LOAD COMB 5 DL LL
107. 1 1.0 2 1.0
108. LOAD COMB 6 DL WX
109. 1 1.0 3 1.0
110. LOAD COMB 7 DL WZ
111. 1 1.0 4 1.0
112. LOAD COMB 8 DL WCOMB
113. 1 1.0 3 1.0 4 1.0
114. LOAD COMB 9 DL LL WX
115. 1 1.0 2 1.0 3 1.0
116. LOAD COMB 10 DL LL WZ
117. 1 1.0 2 1.0 4 1.0
118. LOAD COMB 11 DL LL WCOMB
119. 1 1.0 2 1.0 3 1.0 4 1.0
120. LOAD COMB 12 UDL LL
121. 1 1.4 2 1.6
122. LOAD COMB 13 UDL WX
123. 1 1.0 3 1.4
124. LOAD COMB 14 UDL WZ
125. 1 1.0 4 1.4
126. LOAD COMB 15 UDL WCOMB
127. 1 1.0 3 1.4 4 1.4
128. LOAD COMB 16 UDL LL WX
129. 1 1.2 2 1.2 3 1.2
130. LOAD COMB 17 UDL LL WZ
131. 1 1.2 2 1.2 4 1.2
132. LOAD COMB 18 UDL LL WCOMB
133. 1 1.2 2 1.2 3 1.2 4 1.2
134. PERFORM ANALYSIS

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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 52/ 97/ 4

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

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ORIGINAL/FINAL BAND-WIDTH=    41/    14/    90 DOF
TOTAL PRIMARY LOAD CASES =    4, TOTAL DEGREES OF FREEDOM =    300
SIZE OF STIFFNESS MATRIX =    27 DOUBLE   KILO-WORDS
REQRD/AVAIL. DISK SPACE =    12.4/ 254264.4 MB

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- 135. LOAD LIST 1 TO 4
- 136. PRINT SUPPORT REACTION